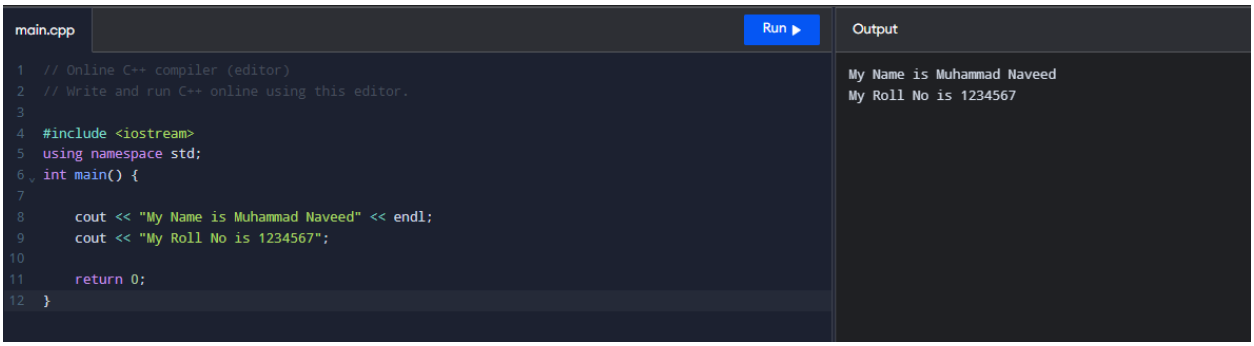
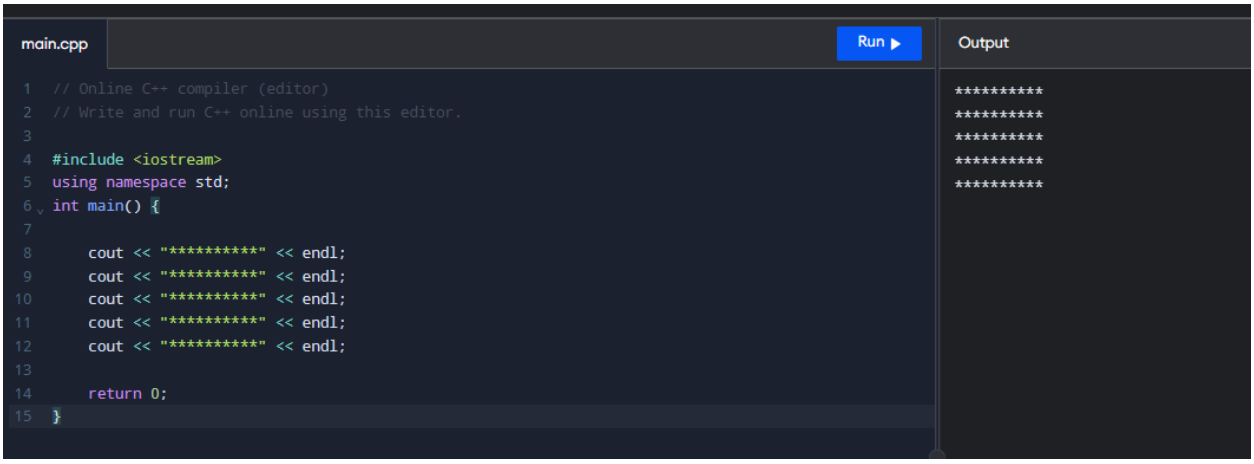
Lab 1:

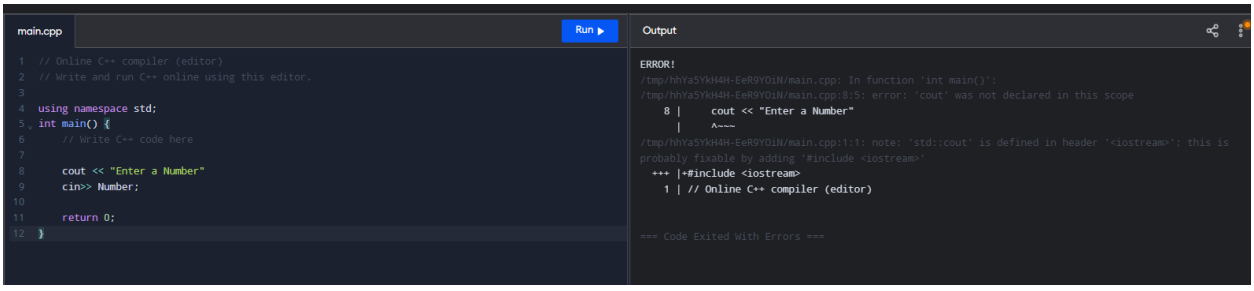
Task 1: Write a Program that prints your name, roll number and Program on separate lines.



Task 2: Print a box of asterisks 5 rows high and 10 columns wide using cout statements.



Task 3: Record three different compiler errors and explain what caused each.



Error 1: Missing iostream Library.

Error 2: Missing Semicolon at the end of cout statement.

Error 3: The variable Named as Number is not decleared.

Task 4: Upload to Github.

Link: https://github.com/MNaveed14122554/PF-Labs

Lab 2:

Task 1: Read the length and width of a field in metres and print its area in square metres and in kanals (1 kanal = 505.857 m²).

main.cpp — Run

```
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;
int main() {
double length, width, area, areaKanal;

    cout << "Enter Length (in Meters): ";
    cin>> length;

    cout << "Enter Width (in Meters): ";
    cin>> width;

    area= length*width;
    areaKanal = area / 505.857;

    cout << "Area is " << area << " Meters" << endl;
    cout << "Area in Kanal is " << areaKanal << " Kanal";

    return 0;
}
```

Output

```
Enter Length (in Meters): 12
Enter Width (in Meters): 34
Area is 408 Meters
Area in Kanal is 0.806552 Kanal
```

Task 2: Read temperature in Celsius and print it in Fahrenheit and Kelvin with two decimal places.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;
int main() {
double celsius, fahrenheit, kelvin;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    fahrenheit = (celsius * 9.0 / 5.0) + 32.0;
    kelvin = celsius + 273.15;

    cout << "Temperature in Fahrenheit: " << fahrenheit << " F" << endl;
    cout << "Temperature in Kelvin: " << kelvin << " K"  << endl;

    return 0;
}
```

Output

```
Enter temperature in Celsius: 37
Temperature in Fahrenheit: 98.6 F
Temperature in Kelvin: 310.15 K
```

Task 3: Read a number of seconds and print it as hours, minutes and seconds using / and %.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;
int main() {
double a;

    cout << "Enter Number in second: ";
    cin >> a;

    cout << a/(60*60)<<" Hours\n";
        cout << a/60<<" Minutes\n";
        cout << a<<" Seconds\n";
    return 0;
}
```

Output

```
Enter Number in second: 60
0.0166667 Hours
1 Minutes
60 Seconds
```

Task 4: Demonstrate overflow by adding 1 to INT_MAX and explain the result.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
#include <climits>   // for INT_MAX

using namespace std;

int main() {
    int max_value = INT_MAX;

    cout << "INT_MAX         = " << max_value << endl;
    cout << "INT_MAX + 1     = " << max_value + 1 << endl;
    cout << "INT_MAX + 2     = " << max_value + 2 << endl;

    return 0;
}
```

Output

```
INT_MAX         = 2147483647
INT_MAX + 1     = -2147483648
INT_MAX + 2     = -2147483647
```

Explanation:

Adding 1 to INT_MAX (2,147,483,647) causes integer overflow because a 32-bit signed int can only hold values up to $2^{31}-1$. In binary, INT_MAX is 0111...1111; adding 1 carries through all bits and flips the sign bit, giving 1000...0000, which in two's complement equals INT_MIN (−2,147,483,648). So the value wraps around from largest positive to smallest negative, like an odometer rolling from 9999 to 0000. (Note: in standard C++, signed overflow is technically *undefined behavior* — the wraparound is just what most compilers do.)

Lab 3:

Task 1: Read three numbers and print the largest using nested if.

main.cpp Run

```cpp

#include <iostream>
using namespace std;
int main() {
int a,b,c;

    cout<<"Enter Three Numbers: ";
    cin>> a>>b>>c;

    if(a>b)
    {

        if(a>c)
        cout<<"Largest Number is: "<<a<<endl;
        else
        cout<<"Largest Number is: "<<c<<endl;

    }

    else if(b>a){

        if(b>c)
        cout<<"Largest Number is: "<<b<<endl;
        else
        cout<<"Largest Number is: "<<c<<endl;

    }
    else
       cout<<"Largest Number is: "<<c<<endl;


cout<< "My name is Muhammad Naveed\nBS.AI\nSection(A)\nMNSUAM";
 return 0;
}


```

Output:

```
Output

Enter Three Numbers: 123 8379 1298
Largest Number is: 8379
My name is Muhammad Naveed
BS.AI
Section (A)
MNSUAM
```

Task 2: Write a grade calculator for marks 0–100 using the grading scheme of the University semester rules.

main.cpp Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int a;
    cout<<"Enter Your Numbers: ";
    cin>> a;

    if(a>=0 && a<=100)
    {
        if(a>=90)
            cout<<"Grade A\n";
        else if(a>=80)
            cout<<"Grade B\n";
        else if(a>=70)
            cout<<"Grade C\n";
        else if(a>=60)
            cout<<"Grade D\n";
        else
            cout<<"Fail\n";
    }

    else
        cout<<"invalid input\n";


cout<< "My name is Muhammad Naveed\nBS.AI\nSection(A)\nMNSUAM";
 return 0;
}
```

Output:

Output

```
Enter Your Marks: 85
Grade B
My name is Muhammad Naveed
BS.AI
Section (A)
MNSUAM
```

Task 3: Write a menu-driven calculator (+, −, ×, ÷) using switch, handling division by zero.

main.cpp Run ▶

```cpp

#include <iostream>
using namespace std;

int main()
{
    double a,b;
    char op;
    cout << "Enter Two Numbers: ";
    cin >> a >> b;
    cout << "Enter Operator (+,-,*,/): ";
    cin >> op;
    switch(op)
        {
            case '+': cout<< "Result of Addition: " << a+b << endl;
            break;
            case '-': cout<< "Result of Subtraction: " << a-b << endl;
            break;
            case '*': cout<< "Result of Multiplication: " << a*b << endl;
            break;
            case '/':
                {
                    if(b != 0)
                        cout<< "Result of Addition: " << a/b << endl;
                    else
                        cout << "Division with 0 is not allowed" << endl;
                }
            break;
            default: cout << "Invalid Operator" << endl;
            break;
        }

   cout << "My name is Muhammad Naveed\nBS.AI\nSection (A)\nMNSUAM\n";

    return 0;
}
```

Output:

```
Output

Enter Two Numbers: 8 3
Enter Operator (+,-,*,/): *
Result of Multiplication: 24
My name is Muhammad Naveed
BS.AI
Section (A)
MNSUAM
```

Task 4: Determine whether a year is a leap year.

main.cpp Run

```cpp

#include <iostream>
using namespace std;

int main()
{
  int year;

    cout << "Enter Year: ";
    cin >> year;

    if(year % 4 == 0)
    {
        cout << "Year is Leap Year" << endl;

    }

    else
        cout << "Year is Not Leap Year" << endl;

   cout << "My name is Muhammad Naveed\nBS.AI\nSection (A)\nMNSUAM\n";

    return 0;
}

```

OutPut:

Output

```
Enter Year: 2026
Year is Not Leap Year
My name is Muhammad Naveed
BS.AI
Section (A)
MNSUAM
```

Lab 4:

Task 1: Print all prime numbers between 1 and 200.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;

int main() {
    cout << "Prime numbers between 1 and 200:" << endl;

    for (int n = 2; n <= 200; n++) {
        bool isPrime = true;

        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0) {
                isPrime = false;
                break;
            }
        }

        if (isPrime)
            cout << n << " ";
    }

    cout << endl;
    return 0;
}
```

Output

```
Prime numbers between 1 and 200:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131
137 139 149 151 157 163 167 173 179 181 191 193 197 199
```

Task 2: Print a right-angled triangle and a pyramid of stars with n rows.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number of rows: ";
    cin >> n;

    cout << "\nRight-Angled Triangle:\n";
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }

    cout << "\nPyramid:\n";
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++)
            cout << " ";
        for (int j = 1; j <= 2 * i - 1; j++)
            cout << "*";
        cout << endl;
    }

    return 0;
}
```

Output

```
Enter number of rows: 12

Right-Angled Triangle:
*
**
***
****
*****
******
*******
********
*********
**********
***********
************

Pyramid:
           *
          ***
         *****
        *******
       *********
      ***********
     *************
    ***************
   *****************
  *******************
 *********************
***********************
```

Task 3: Compute the factorial and the sum of digits of a number.

main.cpp

```cpp
// Online C++ compiler (editor)
// Write and run C++ online using this editor.

#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a number for factorial: ";
    cin >> num;

    long long fact = 1;                 // long long to handle larger results
    for (int i = 1; i <= num; i++)
        fact *= i;

    cout << "Factorial of " << num << " = " << fact << endl;




    return 0;
}
```

Output

```
Enter a number for factorial: 6
Factorial of 6 = 720
```

Task 4: Implement a number-guessing game with at most seven attempts.

main.cpp

```cpp
#include <ctime>      // for time()
using namespace std;

int main() {
    srand(time(0));                  // seed random generator
    int secret = rand() % 100 + 1;     // secret number: 1–100
    int guess;
    const int MAX_ATTEMPTS = 7;
    bool won = false;

    cout << "Guess the number (between 1 and 100).\n";
    cout << "You have " << MAX_ATTEMPTS << " attempts.\n\n";

    for (int attempt = 1; attempt <= MAX_ATTEMPTS; attempt++) {
        cout << "Attempt " << attempt << "/" << MAX_ATTEMPTS
             << " — Enter your guess: ";
        cin >> guess;

        if (guess == secret) {
            cout << "Correct! You guessed it in " << attempt
                 << " attempt(s).\n";
            won = true;
            break;
        }
        else if (guess < secret) {
            cout << "Too low!\n";
        }
        else {
            cout << "Too high!\n";
        }
    }

    if (!won)
        cout << "\nOut of attempts! The number was: " << secret << endl;

    return 0;
}
```

Output

```
Guess the number (between 1 and 100).
You have 7 attempts.

Attempt 1/7 — Enter your guess: 34
Too high!
Attempt 2/7 — Enter your guess: 26
Too low!
Attempt 3/7 — Enter your guess: 29
Too low!
Attempt 4/7 — Enter your guess: 33
🎉 Correct! You guessed it in 4 attempt(s).
```